



# SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

FALL SEMESTER 2026-2027

BCSE497J – Project – I

## GUIDELINES, REVIEW SCHEDULE, EVALUATION RUBRICS

### 1. Course Context and Syllabus Basis

The course aims to provide hands-on learning in the design, development and analysis of a suitable product or process and to strengthen the students' technical skills in the chosen field.

#### Course Outcomes

| CO  | Course Outcome                                                                                       |
|-----|------------------------------------------------------------------------------------------------------|
| CO1 | Demonstrate professional and ethical responsibility.                                                 |
| CO2 | Evaluate evidence to determine and implement best practice.                                          |
| CO3 | Mentor and support peers to achieve excellence in the practice of the discipline.                    |
| CO4 | Work in multidisciplinary teams and provide solutions to problems arising in multidisciplinary work. |

### 2. General Guidelines

- Each project team shall consist of a maximum of three members. The team shall meet the guide regularly and maintain evidence of progress, feedback received and action taken.
- All review comments and suggestions shall be recorded, and the action taken shall be presented during the subsequent review.
- The implementation, results, presentation and report shall reflect the actual work carried out. Each member's contribution shall be clearly demonstrated during the reviews, and every student shall submit an individual final report highlighting their contribution.
- Professional and ethical practices shall be followed in data use, experimentation, citations, code reuse and reporting. Plagiarism, unauthorized copying and undisclosed AI-generated content are not permitted. Any significant use of AI tools shall be appropriately acknowledged.
- The project report shall be prepared progressively, chapter by chapter. The guide shall verify the near-final report during Review 4, and all corrections shall be incorporated before Review 5.

### 3. Review Schedule and Marks Distribution

| Review / Component          | Evaluation Focus                                                                       | Marks      | Review Date(s)     |
|-----------------------------|----------------------------------------------------------------------------------------|------------|--------------------|
| Review 1 (Guide)            | Title, abstract, problem, objectives, scope and preliminary plan                       | 5          | 9-11 July 2026     |
| Review 2 (Panel)            | Domain/problem, literature or patent review, methodology and design                    | 20         | 19 August 2026     |
| Review 3(Panel)             | Follow-up, implementation progress, technical accuracy and interim results             | 20         | 16 September 2026  |
| Review 4(Guide)             | Complete implementation, final results, report verification and final-review readiness | 25         | 12-16 October 2026 |
| Draft/Report Review (Guide) | Cumulative draft report, technical consistency, citations and contribution             | 5          | 12-16 October 2026 |
| Review 5 (Panel)            | Final technical evaluation, demonstration, report, presentation and viva voce          | 25         | 21 October 2026    |
| <b>TOTAL</b>                |                                                                                        | <b>100</b> |                    |

## 4. Evaluation Rubrics with CO Mapping

### Review 1 – 5 Marks

| No. | Evaluation Parameter                    | Marks    | Assessment Focus                                                             | Mapped CO(s)  |
|-----|-----------------------------------------|----------|------------------------------------------------------------------------------|---------------|
| 1   | Project title and technical relevance   | 1        | Specific, meaningful and aligned with the selected domain/problem.           | CO2, CO4      |
| 2   | Abstract and problem definition         | 1        | Clearly states the context, problem and need for the proposed work.          | CO2, CO4      |
| 3   | Project objectives and expected outcome | 1        | Objectives and expected outcomes are clearly defined and achievable.         | CO2, CO4      |
| 4   | Scope and feasibility                   | 1        | Achievable within the semester and available resources.                      | CO1, CO4      |
| 5   | Preliminary work plan                   | 1        | Presents an initial timeline, milestones and allocation of responsibilities. | CO1, CO3, CO4 |
|     | <b>Total</b>                            | <b>5</b> |                                                                              |               |

### Review 2 – 20 Marks

| No. | Evaluation Parameter                                               | Marks     | Assessment Focus                                                                                                                  | Mapped CO(s)  |
|-----|--------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|
| 1   | Domain understanding and problem definition                        | 3         | Understanding of the selected domain; clarity, relevance, assumptions, constraints and significance of the problem.               | CO2           |
| 2   | Literature/patent review and analysis of existing approaches       | 3         | Review of relevant and credible sources; comparison of approaches and identification of limitations or gaps.                      | CO2           |
| 3   | Objectives, scope and expected outcomes                            | 2         | Specific and achievable objectives, clearly defined scope and expected outcomes.                                                  | CO2, CO4      |
| 4   | Proposed methodology                                               | 3         | Appropriateness of methods, algorithms, data, tools, experiments or implementation approach.                                      | CO2, CO4      |
| 5   | System architecture, module design or experimental design          | 3         | Technical soundness of architecture, workflow, module organisation, interfaces or experimental setup.                             | CO2, CO4      |
| 6   | Feasibility, risks, ethics and work planning                       | 3         | Resources, risks, ethical considerations, timeline and allocation of responsibilities.                                            | CO1, CO3, CO4 |
| 7   | Preliminary report quality, presentation and response to questions | 3         | Well-structured report covering Chapters 1-3 at the expected stage; clear presentation, participation and satisfactory responses. | CO3, CO4      |
|     | <b>Total</b>                                                       | <b>20</b> |                                                                                                                                   |               |

### Review 3 – 20 Marks

| No. | Evaluation Parameter                                                         | Marks     | Assessment Focus                                                                                          | Mapped CO(s)  |
|-----|------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------|---------------|
| 1   | Follow-up on previous review feedback and progress against the approved plan | 3         | 1 mark for action taken on Review 2 comments and 2 marks for progress against approved milestones.        | CO1, CO3, CO4 |
| 2   | Implementation and functional progress                                       | 3         | Working modules representing approximately 50% of the approved project scope.                             | CO4           |
| 3   | Technical accuracy and use of best practices                                 | 3         | Correct methods, algorithms, tools, parameters and implementation practices.                              | CO2           |
| 4   | Interim testing, results and analysis                                        | 3         | Preliminary outputs, metrics, tables or graphs with appropriate interpretation.                           | CO2           |
| 5   | Problem-solving and technical refinement                                     | 2         | Identification of technical challenges and evidence of corrective action or improvement.                  | CO2, CO4      |
| 6   | Progress of the project report and documentation                             | 3         | Updated Chapters 1-3 and initial Chapter 4 covering implementation, results and individual contribution.  | CO1, CO4      |
| 7   | Presentation, teamwork/individual responsibility and response to questions   | 3         | Logical presentation, technical understanding, contribution and satisfactory response to panel questions. | CO3, CO4      |
|     | <b>Total</b>                                                                 | <b>20</b> |                                                                                                           |               |

**Review 4 – 25 Marks**

| No. | Evaluation Parameter                                                  | Marks     | Assessment Focus                                                                                                                                       | Mapped CO(s) |
|-----|-----------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1   | Action taken on previous review comments                              | 2         | Evidence that observations and suggestions from Review 3 and the Draft/Report Review have been addressed.                                              | CO1, CO3     |
| 2   | Completion of the approved project scope                              | 3         | Completion of approved modules and deliverables according to project objectives.                                                                       | CO4          |
| 3   | Functionality and integration of modules                              | 3         | End-to-end functioning, proper integration and stable operation of the solution.                                                                       | CO4          |
| 4   | Technical accuracy and quality of implementation                      | 3         | Correctness of methods, algorithms, design choices, tools and implementation practices.                                                                | CO2, CO4     |
| 5   | Testing and validation                                                | 3         | Adequacy of test cases, experiments, validation procedures and supporting evidence.                                                                    | CO2          |
| 6   | Results analysis and interpretation                                   | 2         | Meaningful presentation and interpretation of final results, metrics, tables, graphs or observations.                                                  | CO2          |
| 7   | Technical refinement, limitations and future scope                    | 2         | Improvements/optimisation carried out and clear identification of limitations and future work.                                                         | CO2          |
| 8   | Completeness and technical quality of the final report                | 2         | Near-final report is complete, technically consistent and accurately reflects the work.                                                                | CO1, CO2     |
| 9   | Research publication or intellectual-property submission              | 2         | Valid proof of submission/communication to a peer-reviewed journal, international conference or for patent filing. No marks without documentary proof. | CO1, CO2     |
| 10  | Demonstration, individual contribution and readiness for final review | 3         | Quality of demonstration, clear contribution, incorporation of guide feedback and preparedness for the final panel.                                    | CO3, CO4     |
|     | <b>Total</b>                                                          | <b>25</b> |                                                                                                                                                        |              |

**Review 5 - Final Panel Review – 25 Marks**

| No. | Evaluation Parameter                                                          | Marks     | Assessment Focus                                                                                                 | Mapped CO(s)  |
|-----|-------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------|---------------|
| 1   | Domain knowledge and problem understanding                                    | 4         | Understanding of the domain, problem context, motivation, scope and relevance of the work.                       | CO2           |
| 2   | Methodology, architecture and technical design                                | 4         | Appropriateness and soundness of methodology, algorithms, architecture, modules and design decisions.            | CO2, CO4      |
| 3   | Completion, functionality and integration                                     | 4         | Completion of the approved scope, end-to-end integration and stable functioning.                                 | CO4           |
| 4   | Testing, validation and analysis of results                                   | 4         | Adequacy of testing, metrics, validation, comparison and interpretation of final results.                        | CO2           |
| 5   | Innovation, technical merit and practical relevance                           | 3         | Originality, technical contribution, feasibility, ethical responsibility and practical relevance.                | CO1, CO2, CO4 |
| 6   | Quality of the final report and supporting evidence                           | 3         | Technical completeness, consistency, organisation, references, results, conclusions and individual contribution. | CO1, CO2      |
| 7   | Presentation, demonstration, viva voce and teamwork/individual responsibility | 3         | Clarity, technical understanding, response to questions and contribution of each student.                        | CO3, CO4      |
|     | <b>Total</b>                                                                  | <b>25</b> |                                                                                                                  |               |

**Draft/Report Review – 5 Marks**

| No. | Evaluation Parameter                                          | Marks    | Assessment Focus                                                                                       | Mapped CO(s)  |
|-----|---------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------|---------------|
| 1   | Completeness of report chapters according to project progress | 1        | Required Preliminary Pages and Chapters 1-4 are available at the expected stage.                       | CO1           |
| 2   | Technical consistency between report and implementation       | 1        | Methodology, design, implementation and results in the report match the actual work.                   | CO2           |
| 3   | Quality of analysis, results and supporting evidence          | 1        | Results and claims are supported by relevant outputs, metrics, tables, graphs or observations.         | CO2           |
| 4   | Citations, referencing and similarity compliance              | 1        | Proper citations and references; compliance with the prescribed institutional similarity requirements. | CO1           |
| 5   | Individual contribution and incorporation of guide feedback   | 1        | Contribution is clearly stated and earlier corrections have been incorporated.                         | CO1, CO3, CO4 |
|     | <b>Total</b>                                                  | <b>5</b> |                                                                                                        |               |

## 5. Project Report Structure

| Section                                 | Minimum Content                                                                                                                                                            |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Preliminary Pages                       | Title page; bonafide certificate; declaration; acknowledgement; abstract; table of contents; list of figures; list of tables; list of abbreviations, as applicable.        |
| Chapter 1 - Introduction                | Background and motivation; problem statement; objectives; scope; expected outcomes; organisation of the report.                                                            |
| Chapter 2 - Literature / Patent Review  | Relevant literature and patents, wherever applicable; comparison of existing approaches; limitations; identified gap.                                                      |
| Chapter 3 - Methodology / System Design | Proposed methodology; architecture/system design; module description; algorithms/workflow/experimental design; tools, technologies and datasets.                           |
| Chapter 4 - Implementation and Results  | Implementation details; testing and validation; results and observations; comparison and analysis; limitations.                                                            |
| Chapter 5 - Conclusion and Future Work  | Summary of work completed; major findings; conclusion; limitations and future enhancements.                                                                                |
| References                              | All cited sources in the prescribed referencing style.                                                                                                                     |
| Appendices                              | Additional results, screenshots, test cases, datasets/questionnaires, design/code evidence, publication/patent proof and individual contribution statement, as applicable. |